

TAMIL NADU STATE BOARD - STANDARD 11 CHEMISTRY VOLUME 2

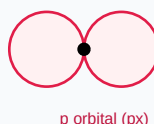
CHAPTER 11: FUNDAMENTALS OF ORGANIC CHEMISTRY

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Chemistry Volume 2
Chapter Name:	Fundamentals of Organic Chemistry

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

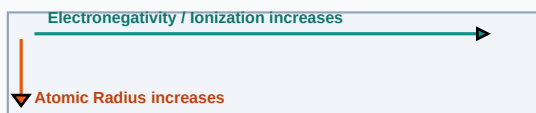
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Catenation	The unique ability of carbon atoms to link together to form long open or closed chain structures.
Functional Group	An atom or group of atoms in an organic molecule that determines its characteristic chemical properties.
Isomerism	The phenomenon where organic compounds have identical molecular formulas but different structural properties.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

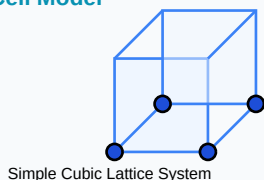
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- IUPAC Name = Prefix + Word Root + Suffix
- Alkanes: $C_n H_{2n+2}$ | Alkenes: $C_n H_{2n}$ | Alkynes: $C_n H_{2n-2}$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

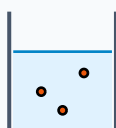
Example 1: Write structural formula of IUPAC name: 2-Methylbut-2-ene.

Step-by-step Solution:

Parent chain is butene (4 carbons with double bond at C2). Methyl substituent at C2.

Formula: $\text{CH}_3 - \text{C}(\text{CH}_3) = \text{CH} - \text{CH}_3$.

Titration Beaker & Solute Precipitation



Reacting Mixture

HOTS (Higher Order Thinking Skills) Challenge

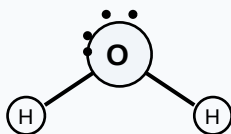
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Classify structural isomerism in organic compounds with clear structures for each category.

Analytical Solution Strategy:

- (1) Chain Isomerism: butane vs isobutane.
- (2) Position Isomerism: propan-1-ol vs propan-2-ol.
- (3) Functional Isomerism: dimethyl ether vs ethanol.
- (4) Metamerism: diethyl ether vs methyl propyl ether.

Lewis Dot Structure (H₂O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

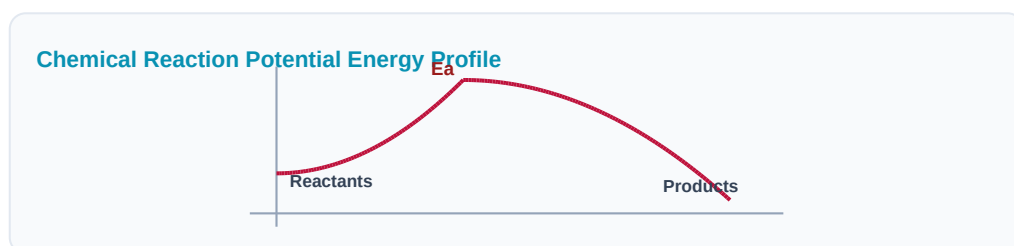
Q1 (1-Mark): Give reasons for the tetravalency of carbon.

Q2 (2-Mark): Write IUPAC names for: (1) $\text{CH}_3\text{-CH(OH)-CH}_3$, (2) $\text{CH}_3\text{-CHO}$.

Q3 (2-Mark): What are homologous series? State their properties.

Q4 (5-Mark): Explain stereoisomerism types.

Q5 (5-Mark): Describe the purification technique of chromatography.



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Petroleum Refining

Distillation cuts isolate linear octane fuels from branched isomer configurations.

Application Case: Perfume Design

Ester functional groups are synthesized to produce sweet, natural fruit fragrances.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

